

### P5.3 Wave interactions

#### Summary

Having studied the electromagnetic spectrum learners now go on to look at the interactions of waves with materials, this will include absorption, refraction and reflection. Learners will also be expected to draw ray diagrams to illustrate the refraction of rays through lenses.

#### Underlying knowledge and understanding

Learners will already be familiar with the properties and behaviour of light. They are expected to have an understanding of behaviour such as reflection, refraction, absorption and scattering. Learners should know that colours are produced by light at different frequencies.

#### Common misconceptions

A common misconception is that when light passes through a coloured filter, the filter will add colour to the light. In addition, learners are often confused about which colours are primary colours.

#### Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers. All other statements will be assessed in both Foundation and Higher Tier papers.

| Topic content                             |                                                                                                                                                 |                                                                                                               | Opportunities to cover: |                                                  | Practical suggestions                                                                                                                                                                                                                                   |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Learning outcomes                         |                                                                                                                                                 | To include                                                                                                    | Maths                   | Working scientifically                           |                                                                                                                                                                                                                                                         |
| P5.3a                                     | recall that different substances may absorb, transmit, refract, or reflect electromagnetic waves in ways that vary with wavelength              |                                                                                                               |                         |                                                  |                                                                                                                                                                                                                                                         |
| P5.3b                                     | explain how some effects are related to differences in the velocity of electromagnetic waves in different substances                            |                                                                                                               |                         |                                                  |                                                                                                                                                                                                                                                         |
| P5.3c <input checked="" type="checkbox"/> | use ray diagrams to illustrate reflection, refraction and the similarities and differences between convex and concave lenses (qualitative only) | how the behaviour of convex and concave lenses determine how they may be used, for example, to correct vision | M5a, M5b                | WS1.1b, WS1.2c, WS1.3a, WS1.3e, WS2a, WS2b, WS2c | Use of concave and convex lenses to investigate how they alter the path of light in different ways. (PAG P4)<br><br>Investigation using convex lenses to see how the image of a light bulb varies with the distance of the bulb from the lens. (PAG P4) |

|                                           |                                                                                                                          |                                    |          |                                                                 |                                                                                     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------|----------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------|
| P5.3d <input checked="" type="checkbox"/> | construct two-dimensional ray diagrams to illustrate reflection and refraction (qualitative only – equations not needed) |                                    | M5a, M5b |                                                                 |                                                                                     |
| P5.3e <input checked="" type="checkbox"/> | explain how colour is related to differential absorption, transmission and reflection                                    | specular reflection and scattering |          | WS1.1b,<br>WS1.2c,<br>WS1.3a,<br>WS1.3e,<br>WS2a,<br>WS2b, WS2c | Use of coloured filters and light sources to investigate how filters work. (PAG P4) |